3.1 Problem Setting

We consider the task of binary classification on four different datasets: modular addition ($x + y$), modular subtraction ($x - y$), modular division (x/y), and permutation. Each dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{\{N\}}$ consists of input-output pairs, where x_i represents the input and y_i the corresponding label.

For the modular arithmetic datasets, we define:

- $x_i = (a_i, b_i)$, where $a_i, b_i \in \{0, 1, \ldots, p-1\}$ and p is a prime number
- $y_i = f(a_i, b_i) \mod p$, where f is the respective arithmetic operation

For the permutation dataset:

- x_i represents a permutation of k elements
- y_i is the result of applying a fixed permutation to x_i

We train a transformer-based model M_θ with parameters θ to minimize the cross-entropy loss:

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_{i=1}^{N} \log P_\theta(y_i|x_i) \tag{1}$$

where $P_\theta(y_i|x_i)$ is the probability assigned by the model to the correct label y_i given input x_i.

To quantify the model's generalization performance, we use validation accuracy. We define the grokking point as the training step at which the validation accuracy reaches 95%.

To estimate the Minimal Description Length (MDL) of the model, we use a weight pruning approach. The MDL at a given training step is approximated by the number of non-zero weights in the model after applying a pruning threshold:

$$\text{MDL}(\theta) \approx |\{w_i \in \theta : |w_i| > \epsilon\}| \tag{2}$$

where ϵ is a small threshold value.

This problem setting allows us to investigate the relationship between MDL, grokking, and generalization across different types of tasks, providing insights into the learning dynamics of neural networks from an information-theoretic perspective.

4 Method

To investigate the relationship between Minimal Description Length (MDL) and grokking in neural networks, we propose a novel MDL estimation technique based on weight pruning. This approach aims to quantify the compression of internal representations during the learning process and relate it to the sudden generalization observed in grokking.

4.1 MDL Estimation Technique

We estimate the MDL of a model with parameters θ by pruning weights below a threshold ϵ and counting the remaining non-zero weights:

$$\text{MDL}(\theta) \approx |\{w_i \in \theta : |w_i| > \epsilon\}| \tag{3}$$

where $\epsilon = 10^{-2}$ in our experiments. This computationally efficient approximation allows us to track changes in MDL throughout the training process.

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4.2 Experimental Setup

We apply our method to the four datasets defined in Section 3: modular addition, subtraction, division, and permutation. For each dataset, we train a transformer-based model Vaswani et al. (2017) with 2 layers, 128 hidden dimensions, and 4 attention heads. We use the AdamW optimizer Loshchilov & Hutter (2017) with a learning rate of 10^{-3}, weight decay of 0.5, and a batch size of 512. Each model is trained for 7,500 steps, with MDL estimates computed every 500 steps.

4.3 Analysis of MDL and Grokking Relationship

To analyze the relationship between MDL and grokking, we introduce several key concepts and metrics:

- **Grokking point**: The training step at which the validation accuracy reaches 95%.
- **MDL transition point**: The step with the steepest decrease in MDL.
- **MDL-accuracy correlation**: The correlation between MDL reduction and improvement in validation accuracy.
- **Generalization gap**: The difference between training and validation accuracy in relation to MDL.
- **MDL transition rate**: The rate of change in MDL over time.

4.4 Visualization and Comparative Analysis

We employ various visualization techniques to compare learning dynamics across datasets:

- Training and validation metrics over time (Figure **??**).
- MDL and validation accuracy combined plots (Figure **??**).
- MDL transition point vs. grokking point scatter plot (Figure **??**).
- MDL-validation accuracy correlation bar plot (Figure **??**).
- MDL evolution and generalization gap plots (Figure **??**).
- MDL transition rate visualization (Figure **??**).
- MDL transition rate vs. grokking speed scatter plot (Figure **??**).

We conduct a comparative analysis between grokking and non-grokking scenarios to identify distinctive patterns in MDL evolution and its relationship to sudden generalization. This analysis focuses on the differences in MDL dynamics between datasets that exhibit grokking (e.g., modular arithmetic tasks) and those that struggle to generalize (e.g., the permutation task).

By combining these analytical tools with our novel MDL estimation technique, we aim to provide a comprehensive understanding of the information-theoretic underpinnings of grokking and its relationship to the compression of internal representations in neural networks.

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5 Experimental Setup

To validate our hypothesis on the relationship between Minimal Description Length (MDL) and grokking, we designed a comprehensive experimental setup to investigate the learning dynamics of neural networks across various tasks. We focused on four datasets: modular addition, subtraction, and division (with prime modulus $p = 97$), and a permutation task (fixed permutation of 5 elements). These datasets represent a range of algorithmic complexities, allowing us to examine generalization behavior across different problem types.

We employed a transformer-based model Vaswani et al. (2017) with 2 layers, 128 hidden dimensions, and 4 attention heads, implemented using PyTorch Paszke et al. (2019). The models were trained using the AdamW optimizer Loshchilov & Hutter (2017) with a learning rate of 10^{-3}, weight decay of 0.5, and a batch size of 512. Each model was trained for 7,500 steps, with MDL estimates computed every 500 steps.

To estimate MDL, we used a weight pruning approach, approximating MDL by the number of non-zero weights after applying a pruning threshold of 10^{-2}. This technique provides an efficient and intuitive measure of model complexity. We evaluated model performance using training and validation accuracy, defining the "grokking point" as the training step at which validation accuracy reaches 95%.

Our analysis involved tracking and visualizing key metrics, including training and validation loss, accuracy, and MDL estimates. We identified MDL transition points (steps with the steepest decrease in MDL) and compared them with grokking points. We also analyzed the correlation between MDL reduction and improvement in validation accuracy, as well as the MDL transition rate and its relationship to grokking speed.

Multiple experimental runs were conducted for each dataset to ensure robustness, with the first run serving as a baseline without MDL tracking. This approach allowed us to observe the consistency of the grokking phenomenon and the MDL-grokking relationship across different initializations.

Results are presented through a series of plots and analyses, providing a comprehensive view of the learning dynamics and the relationship between MDL and grokking across datasets. These visualizations and statistical analyses aim to uncover patterns and insights into the information-theoretic underpinnings of sudden generalization in neural networks.

6 RESULTS

We present the results of our experiments investigating the relationship between Minimal Description Length (MDL) and grokking across four datasets: modular addition (x_plus_y), modular subtraction (x_minus_y), modular division (x_div_y), and permutation. Our experiments used a transformer-based model with 2 layers, 128 hidden dimensions, and 4 attention heads, trained for 7,500 steps using the AdamW optimizer Loshchilov & Hutter (2017) with a learning rate of 10^{-3} and weight decay of 0.5.

Table 1: Final performance metrics across datasets (mean values over 3 runs)

Dataset	Train Loss	Val Loss	Train Acc	Val Acc
x_div_y	0.0054	0.0064	1.0000	1.0000
x_minus_y	0.0146	0.0157	1.0000	0.9998
x_plus_y	0.0054	0.0059	1.0000	1.0000
Permutation	0.0076	5.4155	0.9999	0.3393

Table 1 summarizes the final performance metrics. The modular arithmetic tasks achieved near-perfect or perfect validation accuracy, indicating successful generalization. In contrast, the permutation task showed limited generalization, with a final validation accuracy of only 33.93%.

Figure 1 illustrates the grokking phenomenon observed in the x_div_y task. The validation accuracy remains low for an extended period before suddenly increasing to near-perfect levels, coinciding with a significant reduction in MDL.

Table 2: Grokking points (steps to reach 95% and 99% validation accuracy)

Dataset	95% Val Acc	99% Val Acc
x_div_y	3983	4173
x_minus_y	4403	4610
x_plus_y	2350	2573
Permutation	7347	7390

Table 2 shows the average number of steps required to reach 95% and 99% validation accuracy. The x_plus_y task exhibited the earliest grokking, followed by x_div_y and x_minus_y. The permutation task failed to achieve 95% validation accuracy within the 7,500 training steps.

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